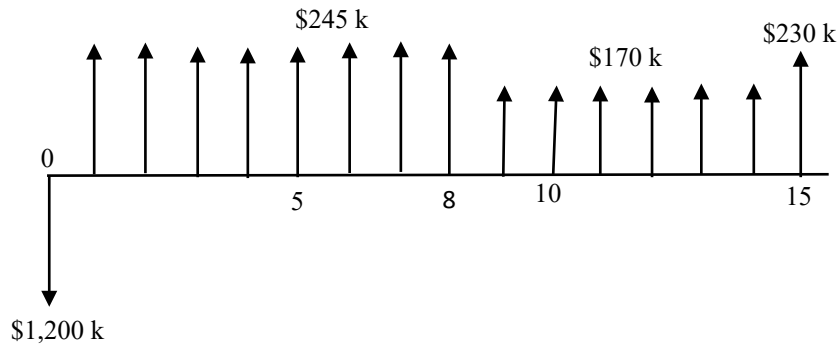


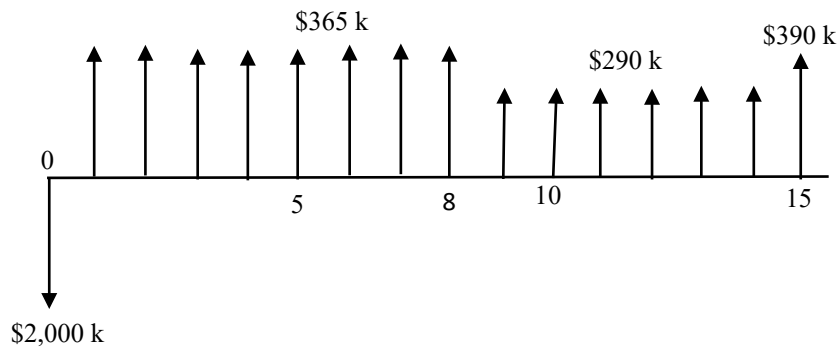
DSS 5202 Sustainable Systems Analysis Solutions to Assignment #2

Question 1 (Cash flow diagrams)

Cash flow diagram for System A



Cash flow diagram for System B.



Question 2 (NPV method)

$$NPV_A(10\%) = \left[-1,200 + \sum_{k=1}^8 \frac{245}{(1+0.1)^k} + \sum_{k=9}^{15} \frac{170}{(1+0.1)^k} + \frac{60}{(1+0.1)^{15}} \right] \times 10^3 \backslash$$

$$= \$ 507,516.50 > 0$$

$$NPV_B(10\%) = \left[-2,000 + \sum_{k=1}^8 \frac{365}{(1+0.1)^k} + \sum_{k=9}^{15} \frac{290}{(1+0.1)^k} + \frac{100}{(1+0.1)^{15}} \right] \times 10^3$$

$$= \$ 629,821.73 > 0$$

$$NPV_B(10\%) > NPV_A(10\%)$$

Hence, System B should be selected based on the NPV method.

Question 3 (*IRR* method)

The incremental analysis method must be used for mutually exclusive alternatives using *IRR*.

IRR of System A is r such that

$$-1,200 + \sum_{k=1}^8 \frac{245}{(1+r)^k} + \sum_{k=9}^{15} \frac{170}{(1+r)^k} + \frac{60}{(1+r)^{15}} = 0$$

Using an equation solver: $r = 17.52\% > MARR = 10\%$
System A is feasible.

IRR of System B is r such that

$$-2,000 + \sum_{k=1}^8 \frac{655}{(1+r)^k} + \sum_{k=9}^{15} \frac{290}{(1+r)^k} + \frac{100}{(1+r)^{15}} = 0$$

Using an equation solver: $r = 15.48\% > MARR = 10\%$
System B is feasible.

Base alternative with lower initial cost = System A

Next alternative = System B

Consider incremental cash flows (B – A)

IRR of (B-A) is r such that

$$-800 + \sum_{k=1}^8 \frac{120}{(1+r)^k} + \sum_{k=9}^{15} \frac{120}{(1+r)^k} + \frac{40}{(1+r)^{15}} = 0$$

Using an equation solver: $r = 12.57\% > MARR = 10\%$
(B – A) is feasible.

Hence, System B should be selected based on the *IRR* method.

Question 4

Comparison of $NPV(10\%)$ and IRR of the two alternatives.

	System A	System B
$NPV(marr)$	\$ 507,516.50	\$ 629,821.73
IRR	17.52%	15.48%

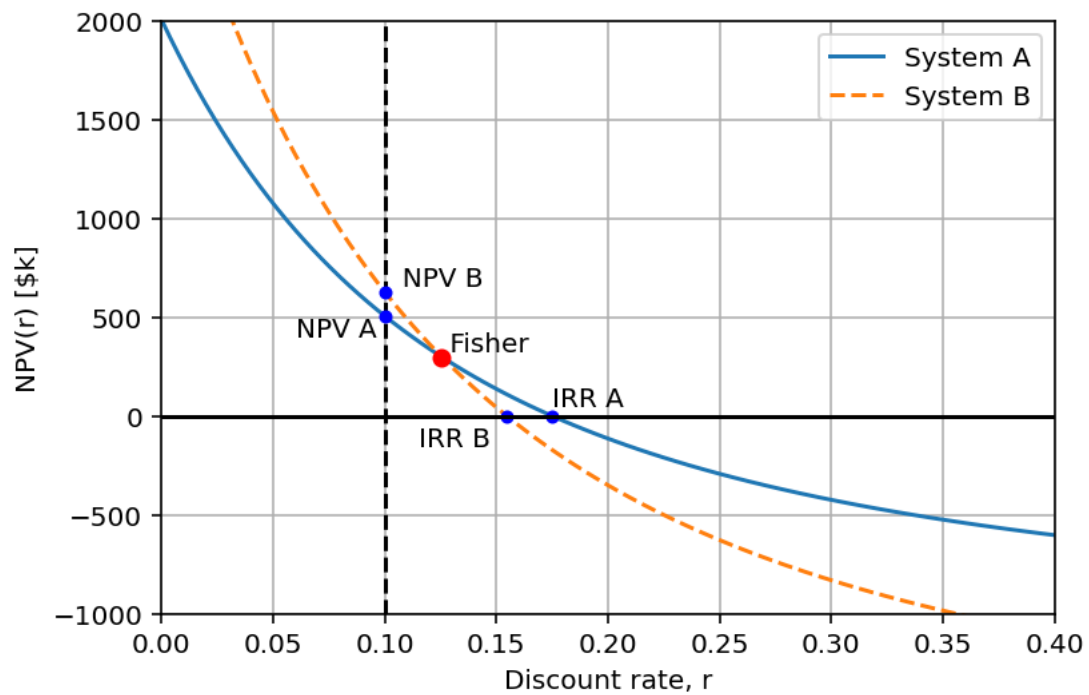
We note that at $MARR=10\%$, NPV of A < NPV of B, but IRR of A > IRR of B.

This is due to the Fisher intersection rate = 12.565% > $MARR = 10\%$

For the ranking by NPV to be the same as the ranking by IRR , we require $MARR \geq$ Fisher rate.

See plot below.

Hence, the statement in the question is true when $MARR \geq 12.565\%$



Question 5 (*MIRR* of System B)

Financing rate = 8%

Reinvestment rate = 10%

MARR = 10%

Negative net cash flows occur in year 0 only:

$$|PV(-ve \text{ net CF at } 8\%)| \\ = - \$ 2,000,000$$

Positive net cash flows occur in years 1 to 15:

$FV(+ve \text{ net CF at } 10\%)$

$$= \left[\sum_{k=1}^8 365 (1 + 0.1)^{15-k} + \sum_{k=9}^{15} 290 (1 + 0.1)^{15-k} + 100 \right] \times 10^3$$

$$= \$10,985,417.99$$

$$MIRR_B = \sqrt[25]{\frac{10,985,417.99}{2,000,000}} - 1$$

$$= 12.026\% > MARR = 10\%$$

Hence, System B is financially feasible based on the *MIRR* method.